

–Lesson plan (Template –2)

Name of the Programme:	BSC(Honors)	
Title of the Course:	Introduction to Version Control	
Course code	CS1105	
Semester:	II & IV	
Names of the Course Instructor(s)	Dr. K Sailaja Kumar (2-Sem) Paras Negi (4-Sem)	
Course credit	No. of hours per week (0:1:2)	Total no. of Teaching hours
02	03 Hours	45 Hours

Session	Topic	Pedagogy /activities	Pre-reading/reference
1.	Introduction to coding platforms: LeetCode, HackerRank, Codeforces	Interactive demo and guided exploration of platforms	LeetCode Handbook, HackerRank Docs, Pro Git Ch.1
2.	Introduction to coding platforms	Register on platforms & solve basic problems	LeetCode Practice, HackerRank Tutorials, Codeforces Guide
3.	Introduction to coding platforms	Continue solving problems; compare solutions	LeetCode Practice, Editorials
4.	Problem-solving strategies on coding platforms	Discussion on strategies, time complexity, patterns	LeetCode Blog, HackerRank Practice Guide
5.	Problem-solving practice	Solve curated problems; analyse logic	LeetCode Discuss, Editorials
6.	Problem-solving practice	Timed problem-solving session	HackerRank Interview Prep
7.	Basics of Version Control: Git & VCS	Centralized vs. Distributed; advantages of Git	Pro Git Ch.1, Git-SCM Docs
8.	Git installation & setup	Install Git; configure user credentials	Git Installation Guide
9.	Git initialization	Create repo; first add & commit	Pro Git Ch.2
10.	Basic Git Commands	git add, commit, status, log	Pro Git Ch.2
11.	Basic workflow	Multi-file commits, explore history	Pro Git Ch.2–3
12.	Inspecting differences	Use git diff; modify/delete files	Git-SCM Docs

13.	Remote Repositories	GitHub overview, local vs remote	GitHub Docs
14.	Push local → remote	Create GitHub repo, push commits	GitHub Quickstart
15.	Pulling & syncing	Pull updates; handle basic conflicts	GitHub Docs
16.	Forking and Cloning	Contribution workflow, fork/clone model	GitHub Collaboration Guide
17.	Forking & cloning practice	Fork repo, clone locally, modify	GitHub Classroom
18.	Pull Requests	Submit PR; review peers' PRs	GitHub Docs
19.	Branching in Git	Create, switch, delete branches	Pro Git Ch.3
20.	Branching practice	Create feature branches; edit code	Git Branching Tutorial
21.	Merging branches	Merge features; observe outputs	Pro Git Ch.3
22.	Merge Conflicts	Why conflicts occur; their resolution	Pro Git Ch.3
23.	Conflict simulation	Trigger merge conflicts	Git Docs
24.	Conflict resolution practice	Resolve conflicts step-by-step	Git Merge Tools
25.	Git Stash	Temporary storage of work-in-progress	Pro Git Ch.6
26.	Stash practice	stash, apply, pop, drop	Git Stash Docs
27.	Stash scenario exercises	Use stash within workflows	Pro Git Ch.6
28.	Git Rebase vs Merge	Linear history, reordering commits	Pro Git Ch.6
29.	Rebase practice	Basic + interactive rebase	Git Rebase Docs
30.	Fixing rebase issues	Squashing, editing commits	Git Tools Docs
31.	Git Tags & Cherry-pick	Release tags, isolated commit transfer	Pro Git Ch.6
32.	Tagging practice	Lightweight & annotated tags	Git Tag Docs
33.	Cherry-pick practice	Pick specific commits into branches	Git SCM Docs

34.	GitHub Co-Pilot Introduction	Installation, setup, ethical overview	Co-Pilot Docs
35.	Co-Pilot coding practice	Generate starter code with AI	Co-Pilot Labs
36.	Debugging with Co-Pilot	Check correctness of suggestions	Co-Pilot Labs
37.	Ethical AI Use & Limitations	Responsible coding with AI	AI Ethics Guidelines
38.	Ethical code review	Compare human vs AI code	Co-Pilot Docs
39.	Documentation & reflection	Write reflective notes on AI use	AI Ethics Docs
40.	Best Git Practices	Commit messages, branch strategy	Git Best Practices Guide
41.	Project setup	Create repos; assign roles	GitHub Project Setup
42.	Group work session	Start project implementation	Team Workflow Docs
43.	Final Integration	Complete Git workflow revision	GitHub Docs
44.	Project demonstration	Students present repositories	SEE Practical Guide
45.	Final assessment & viva	Faculty evaluation & submission	Course Notes

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz Graded Component-1 - IBM SkillsBuild Courses: Graded Component-2	10 marks 10 marks
CIE-2 - Practical Component	25 Marks
CIE-3 - Lab work submission (GitHub) & viva	15 Marks +10 marks

Semester End Examination (SEE): 30%	
Mode of Exam	Practical

Weightage of exam	30 Marks
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Course Outcomes- Programme Outcomes Matrix

Sl. No.	Course outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1.	Understand the purpose, structure, and benefits of centralized and distributed version control systems	3	2	0	2	0	0	0	1	0	0	0	0
2.	Develop problem-solving abilities using coding platforms and real-world programming tasks	3	3	2	2	0	0	0	2	0	0	0	0
3.	Demonstrate proficiency in Git and GitHub for collaborative version control and project management	3	2	0	3	2	2	2	2	0	0	0	0
4.	Apply advanced Git techniques and adopt best practices to enhance workflow efficiency and team productivity	3	3	2	3	2	2	3	3	0	2	0	0

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Quiz Graded Component 1	Quiz will be conducted for 10 marks. Each right answer carries one mark each. No negative marking for wrong answers.
	IBM SkillsBuild Courses: Graded Component-2	Certificate of completion for 10 marks. The specified courses on the IBM SkillsBuild platform to be uploaded to the designated Google Classroom. (Refer Table-1)

Table-1:

Graded Component-2 (5 Marks):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	Excellent (10 Marks)	Satisfactory (≤7 Marks)	Needs Improvement (≤1 Mark)
Completion of Courses (10 Marks)	Successfully completes all specified courses on the IBM SkillsBuild platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Practical Component	The CIE-2 practical examination will be conducted in the laboratory for 25 marks. (Refer Table-2) Writing / Procedure (8 Marks) Execution / Output (12 Marks) Viva-Voce (5 Marks)

Table-2:

CIE-2 (25 Marks): Practical Laboratory Examination

Assess students' practical skills in applying version control concepts using Git and GitHub through hands-on execution and viva-voce.

Criteria	Excellent	Satisfactory	Needs Improvement
Writing the Procedure (8 Marks)	Writes correct Git commands / procedure in proper sequence; shows clear understanding of the task. (8 Marks)	Writes most commands correctly but sequence or explanation has minor errors. (≤5 Marks)	Writes incorrect or incomplete commands; procedure not clear. (≤2 Mark)
Execution and Output (12 Marks)	Executes the task correctly in the system; required Git operations are performed and output is correct. (12 Marks)	Executes most steps correctly; minor execution or output errors. (≤6 Marks)	Unable to execute the task correctly; output is incorrect or missing. (≤3 Mark)
Viva-Voce (5 Marks)	Explains commands, workflow, and output confidently and correctly. (5 Marks)	Gives basic explanation with some hesitation. (≤3 Mark)	Unable to explain commands or workflow. (≤1 Mark)

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab Work Submission (GitHub) & Viva-Voce Total Marks: 25 (15 Marks + 10 Marks) (<i>Refer Table-3</i>)

Table-3:

CIE-3: Lab Evaluation (GitHub Submission with Viva-Voce)

Every laboratory exercise shall be evaluated based on work submitted on GitHub, along with a viva-voce conducted for the same submission.

Criteria	Excellent (2 Marks)	Satisfactory (1 Mark)	Needs Improvement (0.5 Marks)
Completion of Lab Work (2 Marks)	All assigned lab exercises are completed and submitted on GitHub.	Most lab exercises are completed; minor omissions.	Several lab exercises are incomplete or not submitted.

Repository Organization & Documentation (2 Marks)	Repository is well-organized with proper folder structure and basic documentation.	Repository structure is acceptable with limited documentation.	Repository is poorly organized with missing documentation.
Commit History & Version Control Practices (2 Marks)	Regular commits with meaningful messages reflecting proper version control practices.	Few commits or generic commit messages.	Very few or single commits; poor version control practices.
Use of Git Commands & Features (2 Marks)	Appropriate use of Git commands.	Limited use of Git commands.	Incorrect or minimal use of Git commands.
Understanding of Submitted Lab Work (2 Marks)	Clearly explains the submitted lab work, commands used, and workflow followed.	Basic explanation with minor gaps.	Unable to explain the submitted work.

Total Marks		30
Semester End Examination		Rubrics for Assessment
SEE	Practical Component	The SEE practical examination shall be conducted in the laboratory. Evaluation will be based on writing/procedure, execution/output, and viva-voce, with reference to the work submitted on GitHub. (Refer Table-4) Writing / Procedure (10 Marks) Execution / Output (15 Marks) Viva-Voce (5 Marks)

Criteria	Excellent	Satisfactory	Needs Improvement
Writing / Procedure (10 Marks)	Writes correct Git commands and procedure in proper sequence; clearly demonstrates understanding of the given task. (10 Marks)	Writes most commands correctly but with minor errors in sequence or explanation. (≤6 Marks)	Writes incorrect, incomplete, or unclear commands; procedure not properly written. (≤3 Marks)
Execution / Output (15 Marks)	Executes the given practical task correctly on the system; all	Executes the task with minor mistakes;	Unable to execute the task properly; output is incorrect or missing.

	required Git operations are performed and output is correct. (10 Marks)	output is partially correct. (≤ 8 Marks)	(≤ 5 Marks)
Viva-Voce (5 Marks)	Explains commands used, GitHub submission, workflow followed, and output confidently and accurately. (5 Marks)	Provides basic explanation with hesitation or minor conceptual gaps. (≤ 3 Marks)	Unable to explain commands, workflow, or output satisfactorily. (≤ 1 Marks)